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October 16, 1962
Revision of Int. Fed. Spec.
ZZ-P-0051c (COM-NBS)
December 5, 1960

INTERIM FEDERAL SPECIFICATION

PAD, SURGICAL OPERATING

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Washington 25, D.C., based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized Federal agencies to use this Interim Federal Specification as a valid exception to Federal Specification ZZ-P-51, dated May 2, 1933.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Surgical operating pads covered by this specification are primarily intended for obstetrical use in the operating room.

1.2 Classification.

1.2.1 Type, grade and size. Operating pads covered by this specification shall be of the reversible type and of one grade and size.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification.

Federal Specifications:

ZZ-C-450	-	Cloth, Coated (Rubber and Plastic) and Plastic Sheeting (Rolls) for Hospital Use.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-665	-	Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676	-	Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 601 - Rubber; Sampling and Testing.

FSC 6530

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(FOR INSTRUCTIONS FOR OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE)

Military Specifications:

MIL-P-116 - Preservation, Methods of
MIL-L-10547 - Liners, Case, Waterproof

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications forms a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

National Fire Protection Association Publication:

NFPA No. 56 - Safe Practice for Operating Rooms.

3. REQUIREMENTS

3.1 Material.

3.1.1 Coated cloth. The coated cloth used in the manufacture of the pads shall be made from cloth coated on both sides with natural rubber, synthetic rubber, or a mixture of the two. The coated cloth shall be electrically conductive. The rubber compound shall be free from any ingredient which may be harmful to the skin.

3.1.2 Rubber in bulb and tubing. The rubber compound in the bulb and tubing shall be compounded from natural rubber, synthetic rubber, or a mixture of the two, which will meet the requirements of this specification. The rubber compound shall be electrically conductive.

3.1.3 Metal strips. The metal strips used as reinforcements in the apron shall be annealed steel. Strips shall be soft and flexible to permit easy bending after insertion in the apron to allow a trough to be formed for draining purposes.

3.2 Design and dimensions. Design and dimensions shall be as shown in figure 1. The air cushion shall be made from coated cloth not less than 0.030 inch thick. The air cushion rim shall be securely attached to a one piece apron constructed of coated cloth not less than 0.017 inch thick. The full width of the apron shall be reinforced, as shown on figure 1, with metal strips, 3/16 inch wide by 0.032 inch thick (no. 21, Birmingham wire gage) or 1/2 inch wide by 0.016 inch thick (No. 27, Birmingham wire gage). A rubber inflating tube, fitted with a screw-type valve, shall be attached to one side of the air cushion. A rubber bulb with a minimum of 1 inch of rubber tubing of not less than 3/16-inch inside diameter by 1/16-inch wall shall be furnished. Nominal dimensions 3-1/4 inches long by 1-7/8 inches in diameter. One end of the tubing shall be fitted with a threaded connection so that the bulb may be removed from the pad after the cushion has been inflated. The cushion rim shall have a suspension tab of coated cloth with a brass reinforced eye attached at the top. The pads shall be adequately reinforced at the seams, edges and points of attachment so as to pass the pressure test specified in 4.4.4 without leakage.

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3.3 Physical requirements

3.3.1 Coated cloth. The coated cloth shall conform to the requirements for type I, double coated cloth in ZZ-C-450, except as otherwise specified.

3.3.1.1 Electrical conductivity. The electrical conductivity of the coated cloth in the pad and the assembled pad shall meet the requirements of Section 2541 of NFPA Publication No. 56, Safe Practice for Operating Rooms, except resistance maximum shall be 1,000,000 ohms, when tested as specified in 4.4.5.

3.3.2 Bulb and tubing. The bulb and tubing shall not show material hardening or softening and shall not become brittle or tacky after aging as described in 4.4.3.1, after steam sterilization as described in 4.4.3.2, and after immersion in a 5 percent solution of phenol and a 70 percent solution of alcohol as described in 4.4.3.3.1 and 4.4.3.3.2.

3.3.2.1 Electrical conductivity. The electrical resistance of the bulb and tubing shall not exceed 1 megohm when tested as described in 4.4.4.

3.4 Identification marking. Each pad shall be permanently and legibly marked with the name or registered trademark of the manufacturer and the date (month and year) of manufacture.

3.5 Workmanship. Workmanship shall be first class throughout. Pads shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling

4.2.1 Lot. Unles otherwise specified in the invitation for bids, the lot shall be formed as described in Section 6 of Fed. Test Method Std. No. 601.

4.2.2 Sampling for inspection. Unless otherwise specified in the invitation for bids, the sample for inspection shall be taken as described in 4 of Section 6 of Fed. Test Method Std. No. 601.

4.2.3 Sampling for test. Unless otherwise specified in the invitation for bids, the sample for test shall be taken as described in 4.2.3.2.1 of Section 6 of Fed. Test Method Std. No. 601. Sampling plan B of table II, Section 6 of Fed. Test Method Std. No. 601 shall be used.

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4.3 Inspection. The pads shall be examined to determine compliance with all requirements of this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Major:
Pad not complete with all component parts.
Pad not free of cuts, tears, blisters, and holes.
Air cushion not continuously and securely attached to apron.
Air cushion not readily inflatable.
Reinforcing strips not completely and securely sealed.

Minor:
Pad not free of dirt, oil, and other foreign matter.
Metal Reinforcing strips not flexible.
Pad not free of raw edges.
Identification marking not complete, not legible, and not permanently attached.

One major or two minor defects in the inspection sample shall be cause for rejection. Examination shall be restricted to the classified possible defects listed above.

4.4 Tests.

4.4.1 Dimensions.

4.4.1.1 Diameter of tubing. The minimum inside and outside diameter of the tubing shall be determined as described in Method 2221 of Fed. Test Method Std. No. 601.

4.4.1.2 Wall thickness of tubing. The wall thickness of the tubing shall be determined by subtracting the inside diameter from the outside diameter and dividing the results by two. The measurements shall be made as described in Method 2221 of Fed Test Method Std. no. 601, except that (a) the inside and outside diameters shall be determined as near the same point as practicable, (b) a total of 4 measurements (no sets) shall be made, and (c) the smallest of all the values shall be the thickness of the test unit.

4.4.1.3 Drain apron. The dimensions of the drain apron shall be determined as described in Method 2121 of Fed. Test Method Std. No. 601.

4.4.1.4 Diameter of fully inflated pad and air cushion. The diameter of the entire pad when the cushion is inflated and the diameter of the air cushion shall be determined as described in Method 2221 of Fed. Test Method Std. No. 601.

4.4.2 Coated cloth. The coated cloth in the pad shall be tested as required in ZZ-C-450.

4.4.3 Bulb and tubing.

4.4.3.1 Resistance to aging. The resistance of the bulb and tubing to aging shall be determined as described in Method 7221 or 7231 of Fed. Test Method Std. No. 601 except that (a) the temperature shall be 100 degree C. (212 degrees F.), (b) the specimen shall be a ring or strip 1/4 inch wide, (c) two specimens from each test unit shall be tested, and (d) the specimens shall be examined visually only, for hardening, softening, brittleness, and tackiness.

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4.4.3.2 Resistance to steam sterilization. The bulb and tubing shall be subjected to steam sterilization as described in Method 7411 of Fed Test Method Std. No. 601 except that (a) the specimen shall be a ring or strip 1/4 inch wide, (b) two specimens from each test unit shall be tested, and (c) the specimens shall be examined visually only, for hardening, softening, brittleness, and tackiness.

4.4.3.3 Resistance to disinfectants.

4.4.3.3.1 Phenol. The resistance of the bulb and the tubing to a 5-percent solution of phenol shall be determined as described in Method 6711 of Fed. Test Method Std. No. 601 except that (a) the specimen shall be a ring or strip 1/4 inch in width and (b) two specimens from each test unit shall be tested

4.4.3.3.2 Alcohol. The resistance of the bulb and tubing to immersion in a 70 percent solution of alcohol shall be determined as described in 4.4.3.3.1.

4.4.4 Resistance to pressure. Each test unit, selected as described in 4.2.2 or 4.2.3 shall be filled with air under a pressure of 5 pounds per square inch. The pressure shall be maintained for a period of 5 minutes. While under a 5-pound pressure, the test unit shall be examined under water for leakage and any indication of weakness. The number of units tested and the number that showed leakage or any indication of weakness shall be recorded.

4.4.5 Electrical conductivity.

4.4.5.1 Specimen. The specimen shall be the complete pad.

4.4.5.2 Apparatus. The apparatus shall consist of the equipment described in Section 2522 of NFPA Publication No. 56, Safe Practice for Operating Rooms.

4.4.5.3 Procedure. The complete pad with the inflating bulb attached shall be placed on a nonconductive surface. The electrodes shall be not more than 3 feet apart when measurements are made. The readings shall be made within five seconds after the current is applied to the electrode. Measurements shall be made by placing the electrodes at various representative locations in each area as follows:

1. Three measurements with both electrodes placed on cushion.

2. Three measurements with one electrode on cushion and one on apron.

3. Three measurements with both electrodes on apron.

4. Three measurements with one electrode on bulb and one on cushion with tubing in place.

The measurements shall be recorded to the nearest 0.01 megohm below 0.10 megohm; to the nearest 0.05 megohn above 0.010 megohm; to the nearest 0.10 megohm above 0.50 megohm. The measurements in areas 1, 2, and 3 shall be averaged and the average recorded as the conductivity of the cushion and apron (sheeting). The measurements in area 4 shall be averaged and the average recorded as the conductivity of the tubing and bulb (accessories).

4.5 Compliance criteria, civilian agencies. Compliance of the lot with respect to inspection and tests shall be based on the principles described in section 6 of Fed. Test Method Std. No. 601.

5 PREPARATION FOR DELIVERY

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For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1)

5.1.1 Level A. Each pad shall be folded, or rolled to a minimum safe dimension and individually packaged in a close-fitting box conforming to PPP-B-676, or to PPP-B-665, and the box water-proofed in accordance with MIL-P-116, method IC-2.

5.1.2 Level B. Each pad shall be packaged as specified in 5.1.1 except that waterproofing of the boxes shall not be required.

5.1.3 Level C. The pads shall be packaged in accordance with the suppliers standard commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The pads shall be packed in quantities of 24, in boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or to PPP-B-585, class 3. Each box shall have a case liner conforming to MIL-L-10547. Closure and sealing of the liner shall be in accordance with the appendix to the liner specification. The boxes shall be strapped in accordance with the appendix to the applicable box specification. For quantities less than 24, when required (see 6.1), packing shall be as specified herein.

5.2.2 Level B. The pads shall be packed as specified in 5.2.1, except that the containers shall be domestic type and class, and case liners and strapping shall not be required.

5.2.3 Level C. The pads shall be packed to insure carrier acceptance and safe delivery to destination in containers conforming with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or purchase order interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.
(b) Administrative provisions for inspection records (4.1).
(c) Sampling, if other than specified (4.2).
(d) Inspection required (4.3).
(e) Level of packaging and packing required (5.1 and 5.2).
(f) Marking, if other than specified. (5.3).

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples are necessary,

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they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., normally procured to meet Federal Government requirements.

6.4 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:
Rubber goods, NOIBN
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:
Rubber articles, NOI (Specify density)
Truckload minimum weights: -(W) 10.4; (W) 16.4; (W) 21.4 and (W) 30.4 pounds, (W) subject to Rule 34, National Motor Freight Classification.

NOTICE

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